Central Bank Retail Deposits in the Past

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Abstract

European central banks historically offered deposit accounts to individuals and firms other than banks – the closest precedent for retail central bank digital currency. Using novel evidence, we show that these accounts served payment, liquidity management and store-of-value functions. Access to accounts was broad on paper, but narrow in practice due to discretionary introduction criteria, minimum balances, fees, and the absence of remuneration. Under the Gold Standard, central bank retail deposits were often not neutral for monetary policy. During panics, the accounts provided a safe haven. The resulting deposit outflows contracted commercial banks' balance sheets but were not always persistent.

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1 Introduction

Central banks around the world are currently experimenting with or considering issuing central bank digital currency (CBDC) (see, e.g., European Central Bank, 2021; Illes et al., 2025). Retail CBDCs would grant individuals and firms other than banks direct access to central bank reserves (Bank for International Settlements, 2020; International Monetary Fund, 2025; Cecchetti et al., 2026).[1] No matter whether token-, wallet- or account-based, all envisaged designs for retail CBDCs are characterized by a common economic feature: they constitute direct claims of the non-bank public on the central bank, convertible at par into banknotes, and usable as a means of payment.

This paper provides a long-run, comparative perspective on the arguably closest historical precedent for retail CBDC that shared this defining feature: central bank deposit accounts held by retail customers (Bank for International Settlements, 2018; Bindseil, 2019; Bordo, 2021; Bordo and Roberds, 2022). Figure 1 shows that central bank retail deposits made up a non-negligible share of total central bank liabilities in several European countries between 1870 and 1939.[2] It also documents that this share exhibited substantial variation in the cross-section and considerable fluctuation within countries over time.

Although a series of recent papers explores the role of historical central bank retail deposit accounts (Baubeau et al., 2021; Antipa and Rieder, 2026; Jorge-Sotelo, 2026; Mulder, 2026), these studies remain confined to country-specific episodes.[3] The resulting gap in the literature matters because Figure 1 raises many more questions than it answers. Why did note-issuing central banks not in need of retail funding offer deposit accounts to the general public in the first place? What purpose did these accounts serve for retail customers? Who had access in law, and who used them in fact? Which rules governed the accounts? Can these rules explain the cross-sectional and time variation in the relative

[1] A *wholesale* CBDC, by contrast, is restricted to financial intermediaries.

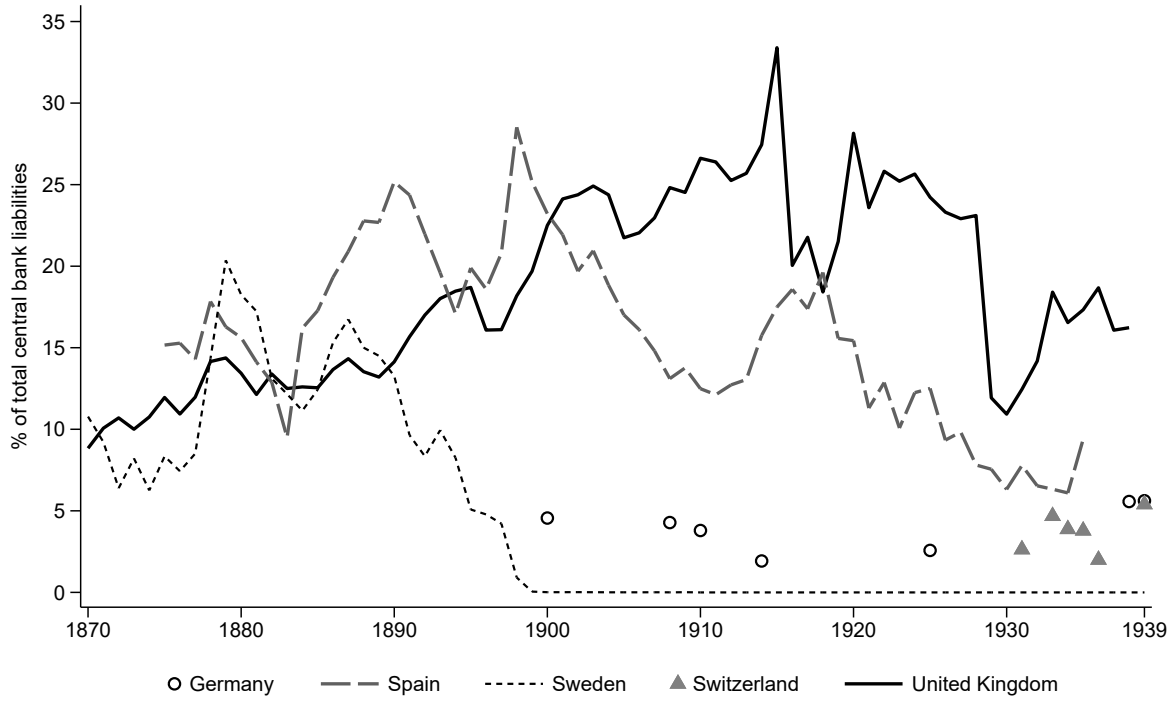
[2] To ensure full comparability, Figure 1 focuses on five selected central banks. We discuss definitions and data availability in Section 2 and Appendix B respectively. We provide additional figures with broader coverage in Appendix A.

[3] In a related paper, Grodecka-Messi and Zhang (2023) study the consequences of the Bank of Canada's absorption of private note issuance after 1935.

importance of central bank retail deposits? Did these deposits at all matter for the conduct of monetary policy in the past? How did these accounts evolve during financial crises? What did flows into central bank retail deposits imply for the commercial banking system?

To address these questions, we gather new quantitative and qualitative archival evidence for a sample of European central banks. We generally cover the time between the date on which these institutions first received the privilege to issue banknotes and the 1930s – when European central banks had largely morphed into the "bankers' bank" and central bank retail deposits had begun to fall out of fashion.[4]

Figure 1: **Retail deposits as % of total central bank liabilities, 1870–1939**



Note: This figure shows the ratio of central bank retail deposits to total central bank liabilities in five selected European countries. We define retail deposits as deposit accounts held by individuals and private sector firms other than banks (including savings banks and other financial institutions). Sources: see Appendix B.

Our sample features the Sveriges Riksbank, the Bank of England, De Nederlandsche Bank, the Oesterreichische Nationalbank (the Austro-Hungarian Bank between 1878 and

[4]The Sveriges Riksbank has issued banknotes since the early 18th century (Brisman, 1918; Fregert, 2018). In this paper, we focus on the period when the central bank offered deposits alongside commercial banks, i.e., from the 19th century onward.

1914), the National Bank of Belgium, the Banque de France, the Banco de España, the German Reichsbank (including its predecessors, the Royal Bank of Prussia and the Prussian Bank) and the Swiss National Bank. Our sample thus covers the central banks of the three key-currency countries during the classical Gold Standard era (Britain, France and Germany) and of most of the smaller open economies that gravitated around them (Flandreau and Jobst, 2005, 2009). It also includes some of the oldest note-issuing institutions in Europe, whose practices served as templates for central banks founded later (Capie et al., 1994; Ugolini, 2017).

Our main take-aways can be summarized in five points. First, just as today's CBDC proposals are motivated by broad policy objectives such as the maintenance of future monetary sovereignty and financial inclusion (e.g., Bank for International Settlements, 2020; Auer et al., 2022; Illes et al., 2025; Cecchetti et al., 2026), retail deposits in the past also served longer-run public goals. Many central banks came to offer retail deposits as a by-product of their lending activities, and then used them to build and operate a national payment infrastructure. Deposit accounts allowed long-distance payments by mere book entry in economies where moving specie or settling bills across cities was slow and costly, fostering the integration of national money markets.

Second, our case studies provide new insight into the supply of and demand for retail central bank deposit accounts in the past. Historically, near-universal access to central bank deposit accounts on paper often coexisted with narrow use in practice. Minimum balances, prevailing non-remuneration, operating fees, discretionary admission, and sparse central bank branch networks screened out many non-financial firms and households. Yet, despite the almost general absence of remuneration, the wider public *did* hold central bank accounts. Demand rested at least partly on the services bundled with these accounts (e.g., cheap long-distance payments and safe storage). In some countries, holding a central bank deposit account was even a prerequisite for access to the central bank's lending facilities, which retail customers could still use at the time. The historical record thus echoes ongoing debates about the attractiveness and eventual take-up of future CBDCs, and how these aspects may vary with specific design features such as

holding limits (e.g., Andolfatto, 2021; Infante et al., 2024; Ree, 2023; Bidder et al., 2024).

Third, we document that central bank retail deposits were generally not neutral for the conduct of monetary policy under the metallic convertibility mandates of the classical Gold Standard era. Price stability was a by-product of the metallic peg (Bloomfield, 1959; Flandreau, 2008; Ugolini, 2016), which central banks sought to defend or restore to achieve convertibility. Specie cover requirements for banknotes enshrined in central banks' statutes served as the main line of defense intended to preserve the fixed parity. Interestingly, central banks' retail deposits were typically exempt from the cover requirements that constrained note issuance – a remnant of the Currency School's narrow conception of money (Hübner, 1854; Fetter, 1965). Hence, inducing the public to hold account balances instead of banknotes mechanically improved cover ratios. Central banks understood this margin of maneuver and, at times, deliberately exploited it. At the same time, retail deposits remained convertible on demand into notes and were therefore contingent claims on the very issuance headroom the cover rules governed (Jobst and Rieder, 2016; Rieder et al., 2023). Central banks were also aware of this constraint and actively managed the joint amount of banknotes and deposits in circulation.

Fourth, we document that central bank deposits regularly served as a safe haven during financial panics, when the public's confidence in the commercial banking system was shaken (cf. Antipa and Rieder, 2026; Baubeau et al., 2021; Jorge-Sotelo, 2026; Mulder, 2026).[5] The central bank only served as the "depositary of last resort" when it was perceived to be safer than private alternatives. A threat to the continuity of the central bank or the sovereign explicitly or implicitly backing it sometimes prevented a flight into central bank accounts (Jobst and Rieder, 2023). Moreover, during financial panics, customers seeking liquid means of payment at times displayed a preference for banknotes over central bank deposits.

Fifth, we document that crisis-driven inflows into central bank deposits translated into a lasting disintermediation of the commercial banking system only under specific condi-

[5]These patterns speak to modern concerns that a CBDC could facilitate system-wide runs from bank deposits into central bank money (Mancini-Griffoli et al., 2018; Fernández-Villaverde et al., 2021; Keister and Monnet, 2022; Williamson, 2022; Tercero-Lucas, 2022; Muñoz and Soons, 2023).

tions. "Ratchet effects", that is, post-crisis persistence of panic inflows, depended on the opportunity cost of holding unremunerated balances at the central bank, on the network externalities offered by the accounts, and on whether the trust in private alternatives had been permanently affected. Moreover, the impact of crisis inflows on the banking system hinged on the channel by which the central bank provided liquid funds to the banking sector to pay out depositors. Where liquidity was offered by way of discounting – an outright asset purchase – rather than by collateralized lending, deposit migration to the central bank could lead to a contraction of commercial banks' balance sheets, violating the conditions underlying the equivalence benchmark of private and public money in Brunnermeier and Niepelt (2019) (see also Fernández-Villaverde et al., 2021; Chiu and Rivadeneyra, 2021; Niepelt, 2021; Williamson, 2023; Niepelt, 2025).

The paper proceeds as follows. Section 2 defines the object of study and delimits its scope. Section 3 describes the use cases of central bank retail deposits in normal times, from the perspective of both the central bank and the public. Section 4 examines access rules, pricing, and the resulting patterns of effective use. Section 5 analyzes the economic implications for monetary policy, financial stability in crises, and competition with the commercial banking system. Section 6 concludes and outlines avenues for future research.

2 Definitions and scope

Contemporaries used different labels for central bank retail deposit accounts. They were called *private drawing accounts* at the Bank of England, *Girokonten* at the Reichsbank, the Oesterreichische Nationalbank (later, the Austro-Hungarian Bank) and the Swiss National Bank, *comptes courants* at the National Bank of Belgium and the Banque de France, *cuentas corrientes de particulares* at the Banco de España, *rekeningen-courant* at De Nederlandsche Bank, and *upp- och avskrivningsräkning* (for demand deposits) or *depositionsräkning* (for time deposits) at the Riksbank. Below, we use the terms "central bank retail deposits" and "current accounts" interchangeably. Throughout this paper, we use the term "central bank retail deposits" to refer to current accounts held by individuals and private sector firms other than banks.

That said, defining and excluding the commercial banking sector from the retail segment is historically not straightforward. Today, the line between banks and non-banks, and between financial and non-financial firms, is drawn by banking law and licensing. During most of the period we study, however, financial intermediation was carried out not only by incorporated banks but also by private bankers, by merchant-bankers trading under their own names and with unlimited liability, by bill brokers and discount houses, and by notaries who matched borrowers and lenders (Pressnell, 1956; García López, 1987; Jonker, 1996; Cassis et al., 2009; Guinnane, 2002; Hoffman et al., 2019). A strict legal category of "banks" in the modern sense did not yet exist. In many countries, it only emerged with the banking laws of the interwar period (Grossman, 2010). As a result, several central banks did not separate banks from firms other than banks in the sources we consulted for this paper. For example, this is the case for the Banque de France or the Oesterreichische Nationalbank. In other cases like Spain, the central bank reported only total private current accounts, but historical estimates of banks' and financial intermediaries' current accounts held with the Banco de España allow for a more precise estimate. Where the distinction between banks and other firms is explicitly made in the sources (for instance at the Bank of England, which differentiated between bankers' balances and other private drawing accounts; or at the Reichsbank and the Swiss National Bank, see Section 4.3), these categories do not necessarily correspond to our modern definition of commercial banks and are not harmonized across countries.[6]

While this lack of harmonization limits the scope for systematic long-run quantitative comparison, it also provides a first substantive insight. The cross-country heterogeneity visible in Figure 1 is not only driven by the accounts' governance details, the economic mechanisms and the key events we discuss below. It likely also stems from different definitions of what counted as a bank in the respective jurisdictions.

Two neighboring instruments fall outside the scope of this paper. First, most central banks offered custodial services for cash and securities, frequently labeled "deposits" in non-English-speaking countries (e.g., *Depositen* in Austria-Hungary, *verschlossene De-*

[6]We provide more details on the original sources in the Appendix and in the replication files.

positen und offene Depots in Germany, *bewaargeving* in the Netherlands, *depósitos de valores nominales* in Spain, *deposita* in Sweden and *Rechnungen der Deponenten* or *Depotguthaben* in Switzerland).[7] These were safekeeping arrangements for non-fungible assets. They performed no monetary function and hence we do not discuss them further.

Second, so-called credit current accounts (e.g., *Kontokorrent* in Austria-Hungary and Germany, *cuentas corrientes de crédito* in Spain, *voorschotten in rekening-courant* in the Netherlands, and advances or Lombard accounts elsewhere) represented collateralized credit lines: holders pre-positioned collateral against which they could draw at interest. Balances drawn on such accounts were operationally similar to deposit balances arising from discount proceeds, but the underlying contract was one of credit, not deposit – a distinction contemporaries drew clearly (e.g., Ureña Smenjaud, 1897) and which was reinforced by the near-universal prohibition of overdrafts on retail current accounts. This prohibition is what kept the deposit contract – the focus of our paper – legally distinct from the credit current accounts, even where the two were operationally intertwined. Retail current accounts were not seen as credit instruments, and central banks policed this boundary. For example, overdrafts were strictly ruled out at the Bank of England, in Austria-Hungary, in France, in Sweden, in Spain and in Germany – where issuing a cheque beyond one's balance could lead to closure of the account and a permanent bar (e.g., Oesterreichisch-ungarische Bank, 1907; Banco de España, 1876; Banque de France, 1880). De Nederlandsche Bank permitted overdrafts only from 1889, and only with explicit authorization and posted collateral – effectively converting the account, at that margin, into a collateralized credit line.[8]

[7] For some periods, the German nomenclature complicates the distinction somewhat as the Prussian Bank called actual retail deposits *Depositen-Kapitale* and safekeeping arrangements *verschlossene Privat-Depositа*.

[8] NL-HaNA, Dutch National Archives (2.25.68), inv. 7146.

3 Use cases in normal times

3.1 The central bank's perspective

Why did central banks offer retail deposits in the past? Unlike commercial banks, a note-issuing central bank does not depend on deposits for funding: it can create its own liabilities at will.[9] The observation that the central banks in our sample nevertheless operated retail deposit facilities therefore requires explanation.

A first important reason for the existence of retail deposit accounts is that they emerged as a by-product of, and a complement to, central bank lending. At the time, individuals and non-financial firms had direct access to the central bank's standing credit facilities. Proceeds from discounted bills or advances were credited to current accounts, and accounts served as the settlement leg of the credit business. In Belgium, this link was constitutive: current accounts existed essentially as by-products of admission to the discount window, and anyone active on the discount market was for this purpose considered a "banker" (Kauch, 1950). At the Banque de France, the conditions for opening a current account were the same as those for admission to the discount window (Banque de France, 1880). Central banks also used the link deliberately to create demand for their accounts: the Austrian National Bank required every discount customer to hold an account and, at times, required that bills of bank customers be made payable at the bank; the Swiss National Bank's internal guidelines of 1907 required counterparties selling it bills to open a current account (Swiss National Bank, 1932, p. 74; p. 86); and during a shortage of small denominations in August 1914, the Swiss National Bank supplied firms with notes and coin for wage payments only if they opened a *Girokonto* and obtained the funds by discounting eligible bills (Landmann, 1914). Similarly, in Germany before World War I, the Reichsbank put pressure on discount customers to set up a deposit account.[10] These

[9] Note issuance during the Gold Standard era was subject to cover requirements, which effectively constrained central banks' ability to print notes. In some countries, inflows into central bank deposit accounts could ease the binding nature of the cover requirement. We discuss the implications of cover rules and retail deposits for the conduct of monetary policy in Section 5.1 below.

[10] Bundesbankarchiv, B332/22 (*Geheime Verfügungen des Reichsbankdirektoriums und Antwortschreiben bzw. Berichte der Reichsbankstelle Emden*).

account relationships gave the central bank access to valuable additional information about its counterparties: observed payment flows and balances informed the calibration of counterparty-specific conditions, from minimum balances to discount limits (Flandreau and Ugolini, 2013) – potentially an early, informal counterpart to the monitoring function discussed in modern CBDC debates (Keister and Monnet, 2022).

From the central bank's perspective, another important rationale for offering retail deposits was to foster an integrated national payment system. In the mid-nineteenth century, national payment systems were fragmented: cash dominated retail transactions, cheques were rare, and transferring money between cities required shipping specie or drawing bills of exchange – processes that were relatively slow, costly, and risky (Nogués-Marco et al., 2019). For example, in Spain, payments between distant cities were settled with eight-day bills of exchange (Castañeda and Tafunell, 1993; Castañeda, 2001). In the Netherlands, long-distance transfers were expensive (in part due to stamp duties) and slow well into the 1890s. The Swiss National Bank, a latecomer founded in 1907, was even explicitly mandated by law to "facilitate payment transactions"; it regarded the payment system as a public good and offered deposits, withdrawals, and transfers free of charge.[11]

Central bank current accounts allowed long-distance payments through mere book entries and thereby contributed to the integration of national goods and money markets, which was particularly convenient in a period dominated by local unit banks in most countries. This function was most fully developed in the giro systems of Germany and Austria-Hungary, where transfers between central bank branches – free of charge for account holders – enabled payments across the entire currency area. The Spanish case illustrates the same dynamic: when the Banco de España expanded its branch network in the 1880s and introduced inter-branch transfers, the relative size of its current accounts more than doubled in less than a decade (see Figure 1). Rapid account growth strained the Bank's ability to screen counterparties. In 1898, after repeated forgeries of signatures on discounted bills, the Banco de España required clients to have discount proceeds credited

[11] Art. 2 of the Swiss National Bank Act of 1905. See Swiss National Bank (1932), p. 319.

to their current accounts rather than paid out in cash, using the account relationship itself as a control device against fraud (Castañeda, 2001).

The payment function of retail current accounts was also facilitated through the introduction of clearing settlement that relied on banks' holding current accounts with the central bank. In Belgium, before the creation of a formal clearing house in Brussels in 1908, transfers between accounts at the National Bank were a principal mechanism of interbank settlement (Kauch, 1950; National Bank of Belgium, 1950). In England, the London clearing bankers all held drawing accounts at the Bank of England and settled their end-of-day positions by transfers across these accounts. The Bank itself joined the London Clearing House in 1864, enabling it to offer its own retail depositors the same transfer velocity as the clearing banks.

In sum, central bank retail deposits served as a complement to central bank credit operations and provided a technology to modernize payments and integrate national markets. These motives also partly explain their life cycle, visible in Figure 1: central banks were often the first institutions with nationwide branch networks, and their accounts placed them at the center of the payment system before large branched commercial banks existed. When commercial banks built their own networks and interbank settlement migrated to clearing arrangements, the relative importance of central bank retail deposits declined – e.g., in Spain visibly from the 1900s, when Banco Hispano Americano and Banco Español de Crédito began to expand nationwide, and in Germany after 1895, when the universal banks pushed into the provinces.

3.2 The end user's perspective

When discussing use cases from the account-holder perspective, it is useful to distinguish normal times from crises. This subsection concentrates on the fundamental, long-run drivers in normal times. Crisis dynamics are treated in Section 5.2.

3.2.1 Payments, liquidity management, store of value and prestige

For retail customers, the accounts served several main purposes in normal times. The first was payments, in particular via access to free or cheap long-distance transfers. Withdrawals could be executed in cash, through transfer instructions or cheques. For example, at the Reichsbank, "white cheques" served cash withdrawals – payable to a named person or bearer – while non-transferable "red cheques" effected cashless giro transfers to named accounts locally or in other cities (Telschow, 1900, pp. 85–110). In Austria-Hungary, tellingly, most cheques were cashed by the account holders themselves rather than passed to third parties (Rauchberg, 1897). Thus, outside the giro transfer business, the accounts seem to have functioned above all as a personal cash-management device vis-à-vis the central bank. Of course, in the case of convertible currencies, account holders retained the option to convert deposits into banknotes or specie at par.

The second main rationale was liquidity management: merchants and firms could concentrate their receipts and payments – including the settlement of bills and the proceeds and repayments of central bank credit – in a single account at the institution where they also discounted when in need of liquid funds at short notice. The tight integration of accounts with the discount window meant that, for commercial customers, the account was less a savings vehicle than the cash leg of an ongoing credit relationship. In addition, central banks had developed a potent infrastructure to cash maturing bills of exchange. In non-crisis times, customers would typically discount bills of exchange at the central bank's standing facility only a few days before maturity. Central banks often had a large network of agents specialized in encashing maturing bills at the bills' acceptors. This service was very convenient for (corporate) customers, as it allowed them to transfer the costs of encashment to the central bank for a small fee (the discount rate).

By contrast, central bank current accounts were rarely used as part of a portfolio with a set of expected nominal returns. With few exceptions (see Section 4.2), balances earned no interest in economies where commercial banks offered strictly positive remuneration on deposits and savings banks catered to small savers at attractive rates. The resulting demand was often correspondingly thin and specific: conservative account holders who

valued safety and liquidity rather than return. In Spain and Britain, contemporaries described central bank accounts as low-risk, low-return assets, attractive mainly because bank deposits were uninsured and individual bank failures could always occur (Antipa and Rieder, 2026; Jorge-Sotelo, 2026). A fully convertible deposit with a central bank on the Gold Standard was arguably the surest way to keep one's gold safe and retrieve it on demand. Customers saw the gold-storage facilities of the central bank as very valuable – both at the retail level, but also at the gross trade level. For instance, London bullion brokers used to store their inventories at the Bank of England (Ugolini, 2013). Still, because an unremunerated account did not even offer an insurance against inflation or depreciation risk, central bank retail deposit accounts did usually not represent an attractive savings vehicle.

Available indicators – high turnover-to-balance ratios and velocity – point to payments and liquidity management, rather than storage, as the primary use cases. Account holders treated the accounts as payment rails and minimized idle balances. At the Reichsbank, the average sojourn time of a deposited mark fell from three days in 1876 to less than one day by 1907 (Reichsbank, 1925, pp. 48–49). Similarly, before the Great Depression, each Swiss franc was transferred more than once each day on average (Swiss National Bank, 1957, p. 273). The Austro-Hungarian Bank's leadership complained repeatedly that, while its discounting, encashment and transfer services were much appreciated, "the willingness to hold deposits was low" (Rauchberg, 1897).

Exceptions proved the rule. In rare cases, central bank deposits were remunerated or enjoyed additional guarantees. At the Royal Bank of Prussia, the King formally guaranteed deposits to address the lack of initial confidence in a bank of issue under government control. Deposits initially paid up to 3%, offering an attractive return in an economy where trustworthy alternatives were scarce. This remuneration may also have helped to overcome depositors' initial reservations. Over time, the interest rate even had to be reduced to 2%, as the safety provided by the explicit state guarantee made 3% too attractive (Niebuhr, 1854, pp. 77–78). The Prussian Bank's deposits retained their store-

of-value role even after the royal guarantee officially lapsed.[12] In Sweden, the Riksbank's interest-bearing accounts (1855–1899) attracted a broad clientele because of their function as savings vehicles (see Section 4.3). The Bank of England never remunerated drawing accounts but it introduced the possibility to grant preferential discount rates for drawing-account customers starting in 1878 – a potential *de facto* remuneration by indirect means, and an early example of rewarding account holding through bundled services rather than interest.[13]

Finally, clients holding an account with the central bank could and did exploit that fact as a device to signal prestige. German companies that held an account with the Reichsbank, for instance, printed this information on their letterheads and routinely published it in financial yearbooks.

3.2.2 Crediting central bank retail deposit accounts

To enjoy the advantages of current accounts at the central bank, retail customers first needed to credit them. Central bank retail accounts could be credited through several channels: paying in banknotes or specie at the counter; crediting cheques drawn on other accounts; depositing bills about to mature or receiving financial income such as dividends and coupons; and, importantly, via the proceeds of discount or Lombard operations.

Figure 2 illustrates the various crediting processes and their impact on the central bank balance sheet. Panel A shows the case of an account holder crediting its current account by depositing central bank notes. Central bank current accounts were a call liability convertible to central bank notes at par, so this type of operation did not affect the size of the balance sheet and just changed the composition of the liabilities. The composition of liabilities, however, could have an impact on the cover ratio – a point directly related to the conduct of monetary policy during the Gold Standard era discussed in Section 5.1.

Panel B in Figure 2 shows the effect of crediting a current account with specie (gold or silver coins, or gold-convertible currencies). No matter whether the central bank operated

[12] After 1846, the guarantee was limited to religious endowments and no longer covered the broad public.

[13] While drawing account holders benefited from preferential discount rates at the Bank, these rates may still have been above the prevailing market rate.

Figure 2: **Crediting central bank retail deposit accounts**

Panel A. Cash deposit

Assets	Liabilities
	+ Current accounts − Banknotes in circulation

Panel B. Specie deposit

Assets	Liabilities
+ Specie reserve	+ Current accounts

Panel C. Lending operation

Assets	Liabilities
+ Bill of exchange or advance	+ Current accounts

Panel D. Cashing a cheque or direct transfer

Assets	Liabilities
	+ Current account (payee) − Current account (payer)

Note: This figure shows the different options available to retail customers to credit their current accounts held at the central bank.

a convertible or an inconvertible currency, funding the current account with specie *did* change the size of the balance sheet. In countries with gold-convertible currencies, this type of operation would increase the total assets of the central bank by increasing its gold reserve, and at the same time push up its liabilities by crediting the current account. The impact on the size of the balance sheet was the same in countries operating inconvertible currencies, like Spain; there, however, no fixed exchange rate existed between gold and central bank liabilities. Hence, in countries with inconvertible currencies, as coin paid in would have been credited at its nominal value, not at its higher market value, funding processes via specie virtually never happened. Whenever individuals stored gold with the central bank, they would do so by using custodial services rather than current accounts (Jorge-Sotelo, 2026).

Current accounts could also be credited when the central bank issued a short-term loan (see Panel C in Figure 2). In the nineteenth century, central banks' most common credit operation was the discount loan, that is, the outright purchase of short-term com-

mercial assets (bills of exchange with a remaining maturity up to 90 days) at a "discount" (the interest rate). In the case of most central banks, access to discount facilities was closely tied to holding a current account, so that the proceeds of discounting could be credited directly to the client's balance. This process caused both the assets and the liabilities of the central bank to increase. The client could then choose to either draw down the proceeds in cash or to retain the balance for later payment purposes, including via cheques. During the twentieth century, discounting was gradually replaced by advances, that is, collateralized lending operations (Jobst and Ugolini, 2016). Advances had the same implications for the central bank balance sheet as discounts.

To fund their account, retail customers could also bring in cheques payable by another current account holder at the central bank. Provided that the credited account holder did not withdraw cash, the operation left the size and composition of total central bank liabilities unchanged, as it simply reallocated deposits across accounts (see Panel D in Figure 2).

4 Access, design and governance

Universal access is an essential dimension of the modern CBDC debate. Its historical counterpart is the question of who could open a central bank retail deposit account, on what terms – and who actually did. Across our cases, governance arrangements reveal a recurring tension: central banks sought to limit the costs and risks of a business they regarded as ancillary, while preserving its usefulness for payments and for their lending operations. The tools they used – admission rules, pricing, and balance requirements – jointly determined both formal and effective access. Table 1 summarizes the design features documented in the previous section and in this section for each central bank in our sample. Below, we discuss the formal rules and the room for discretion first (Section 4.1), then turn to the pricing structures (Section 4.2), before delving into the resulting patterns of effective use (Section 4.3).

Table 1: Design and governance of central bank retail deposit accounts after 1850

Institution	Account name (original)	Remuneration	Fees	Minimum balance	Holding limits / caps	Overdrafts	Deposits in cover requirement?
Austro-Hungarian Bank	*Girokonten*	None	Transfers free; encashment fees on non-discounted instruments (1856)	Function of annual turnover, adjusted with usage	None	Prohibited	No (but contingent notes under pre-1888 ceiling)
Banco de España	*Cuentas corrientes de particulares*	None (option retained; never used)	None documented	Fixed: 2,500 / 1,000 ptas first deposit (Madrid/branches); 250/125 subsequent deposits	None	Prohibited	No (outside note-circulation cap)
Banque de France	*Comptes courants*	None	None; fee on some inter-branch transfers	None	None	Prohibited	No; from 1928: 35% gold cover on notes and current accounts
Bank of England	*Drawing accounts*	None; preferential discount rates for account holders from 1878	None	No posted sum; account had to be "remunerative" (activity-based)	None	Prohibited	No (1844 Act silent on deposits)
De Nederlandsche Bank	*Rekeningen-courant*	None	0.5 ‰ per deposit (until 1914); withdrawals free	None	None	From 1889, with authorization and collateral	Yes: 40% metallic cover on all sight liabilities
National Bank of Belgium	*Comptes courants*	None (statutory prohibition)	None documented	None (access via admission to discounting)	None	Not permitted	Yes (minimum bullion coverage fixed at 33% of all sight liabilities)
Prussian Bank	*Depositen-Kapitale*	2% (2.5% for deposits of minors)	None	50 thalers minimum deposit	Remuneration capped at 5,000 thalers (after 1851)	Prohibited	No
Reichsbank	*Girokonten*	None (interest-bearing deposits ended by 1879)	None direct; fee on payments by non-account holders from 1884	Individual, activity-based; centralized formula from 1906	None	Prohibited	No
Sveriges Riksbank	*Upp- och avskrivningsräkning* (demand); *depositionsräkning* (time)	1855–1899, by holder type; none thereafter (1897 Act)	None documented	None documented	Remunerated only up to tiered caps: 75,000 kr individuals (later 10,000); 300,000 kr banks; 1.2m kr Stockholms Enskilda Bank	Not possible	No
Swiss National Bank	*Girokonti*	None (exception: Confederation)	None (all payment services free)	Individual, with various changes from 1907 to 1946: c. 1% of capital (banks), ≤10,000 CHF (persons)	1909–11, 10-day term to close an account; then none	Prohibited; banks in the clearing system could draw below minimum balances during the day	1907–11: full cover in high-quality assets (Art. 21); then none

Sources: see text and Appendix B.

4.1 Formal access rules

Because most central banks in our sample were private or quasi-public institutions – some of which had evolved out of commercial banks with pre-existing deposit business – access to current accounts was rarely governed by a distinct legal category comparable to prospective CBDC legislation. Instead, access conditions emerged from the interaction of general contract law, the institution's statutes, and its internal rules.[14]

Formally, eligibility was cast in universal or near-universal terms almost everywhere. De Nederlandsche Bank – barred from taking deposits at its foundation in 1814 to protect the Amsterdam *kassiers*, a prohibition lifted at the charter renewal of 1838 – offered accounts to "any person, firm or institution known to the Bank or properly introduced," with no minimum deposit or balance requirement (Kymmell, 1992).[15] The Austro-Hungarian Bank admitted anyone of "undoubted reputation and solidity" expected to maintain a balance "commensurate with [their] circumstances and requirements" (Oesterreichisch-ungarische Bank, 1907, p. 262). The Swiss National Bank opened accounts to "everyone, i.e., banks, other firms, public entities and private persons," and eventually even to foreign firms trading regularly with Switzerland (Swiss National Bank, 1932, 1957, p. 316 and p. 267, respectively). In Prussia, any adult could open an account at the Royal Bank, and rejections were uncommon (Niebuhr, 1854); broad formal access persisted under the Prussian Bank and the Reichsbank even after the state guarantee and the obligation to accept deposits lapsed.[16] In Sweden, the interest-bearing accounts introduced in 1855–1858 were open to the public at large – and after 1874 banks were excluded from them, making these facilities retail by construction (Brisman, 1931; Fregert, 2018).

Formal openness, however, was everywhere combined with discretion. As with the eligibility of bills at the discount window, admission rested partly on deliberately vague

[14] Sveriges Riksbank stands out as a central bank that has been public since its creation in 1668 (Grodecka-Messi et al., 2025). As the only bank in the country, though, for a long time it conducted operations traditionally attributed to commercial banks. Only in the 1830s did competition with commercial banks emerge (Grodecka-Messi and Zhang, 2025).

[15] Dutch National Archives (NL-HaNA), De Nederlandsche Bank (2.25.68), inv. 16431: terms and conditions and name list of current account holders, 1913. Women could open accounts, albeit through an administrator.

[16] Preussische Bank-Ordnung vom 5. Oktober 1846, §§21–28.

criteria – "trustworthiness," "respectability," "solidity" – that allowed the institution to shape its clientele. For example, applicants to the Bank of England had to be "respectably introduced" to the Court of Directors (Clapham, 1958; Hankey, 1867). The Banque de France opened accounts to any person or firm on the same conditions as for admission to the discount window: a written application to the Governor, accompanied by the signatures of three known persons who vouched for the applicant's signature and probity (Banque de France, 1880). The same was true in Belgium, where the legitimacy of deposits was stated exclusively in terms of their complementarity to discounts (Kauch, 1950). The Reichsbank conditioned admission on "absolute trust," a deliberately undefined standard that permitted rejection without explanation. In cases involving members of the Polish minority, a confidential directive from 1902 instructed branch managers to avoid written refusals wherever possible and to communicate rejections orally, in order to prevent political exploitation.[17] The Reichsbank could also revise conditions or close accounts unilaterally (Telschow, 1900). The Banco de España opened accounts to "individuals or companies who ask for it and meet the conditions that the Governing Board decides," excluded undischarged bankrupts and persons with a criminal record, and relied on introduction by existing clients to address remaining information asymmetries.[18] The Swiss National Bank could refuse or close accounts without justification – and, conversely, could compel counterparties to open them to access its lending facilities (Swiss National Bank, 1957, p. 270).

4.2 Pricing: remuneration, fees, and balance requirements

A striking common feature of many historical central bank retail deposits is the absence of remuneration. With few exceptions, balances earned no interest – in stark contrast to contemporary commercial practice. In some countries, remuneration was only phased out over time. Breaking with the tradition of its Prussian predecessors, the Reichsbank terminated all interest-bearing deposits by 1879 after the Bank Act of 1875 capped their

[17] Bundesbankarchiv, B332/8727 (confidential directive of January 1902).
[18] Banco de España (1876, 1874).

acceptance, pivoting entirely to unremunerated giro accounts (Reichsbank, 1901, p. 51).[19] The Riksbank had remunerated demand and time deposits from the 1850s until 1897, but was obliged to wind down interest-bearing accounts after it had received the note monopoly: interest payments ceased in 1899 (Brisman, 1931; Wetterberg, 2010). Spain represented a special case on paper as the statutes granted the Board of Governors the option to remunerate current accounts "in very specific circumstances." However, this option was neither publicized nor used (Banco de España, 1874; Jorge-Sotelo, 2026).

Zero remuneration did not translate into zero pricing. Central banks recovered the costs of running retail deposit accounts through three mechanisms, in varying combinations. The first was explicit fees. De Nederlandsche Bank charged a commission of 0.5 ‰ per deposit – applied also to incoming inter-account transfers until 1904, and abolished altogether on 31 July 1914 – while withdrawals and outgoing transfers were free. This transaction-cost screen discouraged small, low-frequency users.[20] The Austro-Hungarian Bank's encashment fees (i.e., fees for crediting the current account), introduced in 1856, depressed account turnover for decades – an early illustration of how sensitive usage was to transaction pricing (Bubenik, 1888). The Swiss National Bank, an exception, charged nothing at all, treating free payments as part of its statutory mandate (Swiss National Bank, 1932, p. 319). While the Banque de France's current accounts were likewise free of charge and came with a free cheque book, transfers to accounts at other branches were only free if the funds stemmed from a discount or an advance obtained from the Bank (Théry, 1895, p. 46).

The second mechanism, and the workhorse of the system, was the minimum balance – best understood as an implicit fee calibrated to expected account activity. The Austro-Hungarian Bank calculated the required minimum as a function of each account's annual turnover and adjusted it whenever turnover changed: since transfers were free and balances uncompensated, the idle balance was the price of the payment services consumed (Oesterreichisch-ungarische Bank, 1907, p. 270). The Reichsbank operated the

[19] Nevertheless, there is evidence that the central bank paid interest on deposit accounts during the interwar period (Bopp, 1953, p. 459).

[20] NL-HaNA, Dutch National Archives (2.25.68), inv. 16431 and 16423.

same model: minimum balance rules were never publicized but individually negotiated, initially at branch discretion, and from 1906 centrally calculated in Berlin on the basis of each client's transfer volume and the profitability of their other business with the central bank. The effective tightening in 1906 backfired because clients learned to optimize holdings down to the bare minimum, and total non-public balances fell even as required minima rose (Reichsbank, 1925, pp. 48–49). The Bank of England posted no fixed minimum at all: the only condition was that an account be "remunerative," meaning that its ordinary balance had to allow the Bank, "by the use of the balance at the average rate of interest for money, to realize a profit greater than the expense of keeping the account" – with "no stipulated sum insisted upon as a cash balance" (Hankey, 1867, p. 101). The required balance thus varied with the number of cheques drawn and the work the account generated: the same activity-based implicit pricing as in Vienna and Berlin, administered case by case. The Swiss National Bank set individual minimum balances from 1907 to 1946 – as a guideline, 1% of share capital for banks and at most 10,000 francs for private persons.[21] Only Spain relied on fixed, posted thresholds: an initial deposit above 2,500 pesetas in Madrid (1,000 at branches) with subsequent contributions above 250 (125) pesetas, unchanged in nominal terms from the 1870s to the 1930s (Banco de España, 1874, 1876).

The third mechanism operated through caps and tiering. These arrangements are the closest historical analogues to the tiered-remuneration and holding-limit proposals in today's CBDC design debate (Infante et al., 2024). Before 1897, the Riksbank paid interest on demand deposits only up to ceilings defined separately for each holder type – 75,000 kronor for private individuals (later reduced to 25,000 and then 10,000 as the business proved loss-making), 300,000 for private banks, and 1.2 million for Stockholms Enskilda Bank in view of its clearing functions (Brisman, 1931). The Prussian Bank capped remuneration at 5,000 thalers after 1851, accepting larger balances without interest. The

[21] These minima were barely binding for large banks, and were abolished in 1946 (Swiss National Bank, 1957, p. 268). Internal minutes of the Swiss National Bank Directorate and Bankausschuss accessible at the Swiss National Bank's archive show that the minima were often renegotiated by banks and larger firms. These minutes also show that for retail customers minimum balances were regarded as making opening an account at the Swiss National Bank unattractive.

Reichsbank's minimum balances had a second, strategic dimension pointing in the same direction: beyond screening out unprofitable retail clients, they extracted blocked balances – between a quarter and two-fifths of all non-public giro deposits in 1898–1912 – from the large banks that used its payment infrastructure, functioning as a quasi-reserve requirement decades before formal reserve regulation (Deutsche Bundesbank, 1976, p. 42).

Taken together, the governance of central bank retail deposits combined non-remuneration with fees, activity-based balance requirements, or institutional restrictions on eligible counterparties. The different combinations of these tools solved a common challenge: recovering the costs of a service the public expected to be free, without competing too visibly with private intermediaries (see Section 5.3).

4.3 Effective access and clientele

If formal rules projected openness, effective use was narrow nearly everywhere. The interaction of pricing with the prevailing income and wealth distribution, and with the geography of branch networks, meant that central bank accounts primarily served firms, merchants, public bodies, and wealthy households rather than ordinary wage earners. The central question, therefore, is not who was legally able to open an account but who had the "right name" and contacts to get admitted, and who could bear the cost of opening or using an account.

Where thresholds were fixed and posted, the sorting was starkest. The Banco de España's opening requirement of 2,500 pesetas in Madrid, constant in nominal terms for six decades, still amounted in 1930 to roughly 1.7 times Spanish GDP per capita. This sum was the equivalent of about sixteen months of a Barcelona industrial worker's earnings, and some 5 times the balance of the average savings-bank client, as the median depositor at commercial banks held only between 400 and 600 pesetas. Of the nearly 2,000 new current accounts opened at the central bank's Madrid headquarters during the 1931 crisis, less than 1% belonged to day laborers (Jorge-Sotelo, 2026).

Where minima were individualized, the outcome was similar. The lowest balance tolerated by Austro-Hungarian Bank officers in 1907 – 500 crowns – corresponded to

about a fifth of the annual salary of a low-ranking bank employee; on the eve of the First World War the bank maintained 5,761 current accounts – a tiny number in comparison to the 392,000 checking accounts held at the state-guaranteed Austrian Postal Savings Bank (Oesterreichisch-ungarische Bank, 1907). Expert witnesses before the 1908 Reichsbank Inquiry reported that several small manufacturers and tradesmen could not think of establishing a direct connection with the Reichsbank, since the burden of keeping the minimum deposit demanded by the Bank's officials seemed prohibitive to them (National Monetary Commission, 1910, p. 978); and the discretionary trust criterion added social and political selectivity on top (Blum, 1896).

The composition of accounts and balances quantifies this selectivity. At the Reichsbank in 1900, industry and commerce supplied nearly three-quarters of non-public accounts but only 36% of balances, while the banking sector held 59% of balances (70% by 1914) from 18% of accounts. By 1914, the largest 460 accounts held 71% of all non-public giro deposits (Reichsbank, 1925, p. 52). In Switzerland, retail customers (trade and manufacturing firms) held 57% of the National Bank's 1,277 accounts in 1931 but only 8% of balances. The average retail account at the Swiss National Bank held 95,000 francs, a large amount compared to the mean account holdings of 3,300 francs at the postal giro (Swiss National Bank, 1932, p. 324). Bank of England account holders were similarly select: the average daily balance on individual accounts in 1866, about £850, amounted to roughly half the annual income of the best-paid professions (Antipa and Rieder, 2026). For France, no comparable micro-level evidence on account holders is available. The Banque de France imposed neither fees nor minimum balances. Yet, when the renewal of its charter was debated in the 1890s, contemporaries complained that its services did not benefit small savers and firms but were geared towards the needs of the *haute finance* (Théry, 1895, p. 3). Only towards the end of the nineteenth century did smaller firms increasingly start to use its deposit and discount facilities (Théry, 1895, p. 19).

Three further mechanisms of effective exclusion deserve emphasis because they operated without any explicit threshold. The first was transaction pricing: in the Nether-

lands, the absence of a minimum balance was offset by the per-deposit commission and non-remuneration, which made accounts unattractive for small users. The clientele that opened accounts in the first year of the business consisted of the Bank's president, four banking houses, a notary, and six private depositors.

Geography was the second exclusion mechanism. In Prussia, where pecuniary barriers were low and deposits even remunerated, access ran through a handful of *Komtoirs* (six around 1800, eleven by 1875), so that distance to these branches itself rationed access. In Sweden before 1868, customers outside the head office cities had to deposit through intermediaries, often incurring several days of delay. The Swiss National Bank maintained some 340 banking places, including agencies operated by private and cantonal banks, against more than 4,000 post offices, and Swiss customers could not choose their banking location, which was assigned by region (although some exceptions occurred).

The third mechanism was the *de facto* outsourcing of the retail segment to other providers. The Swiss postal giro offered small minimum balances, remuneration up to a cap, and ubiquitous access. Since the central bank and the Post deliberately integrated their systems (free mutual transfers from 1908–09 on) rather than competing, they entered into a division of labor.[22] The central bank served the wholesale payment layer and the Post the retail layer (Swiss National Bank, 1957, p. 270). Similarly, the National Bank of Belgium was designed from its foundation in 1850 not to conduct a retail deposit business: only accounts serving the discount and settlement business were envisaged, and the central bank was statutorily barred from paying interest on its liabilities (Section 5.3 discusses the origins of this design). Access to retail accounts was merely a by-product of admission to the discount window. Since ordinary households neither held eligible commercial paper nor participated in the discount market, they were excluded functionally, without any explicit prohibition on persons (Kauch, 1950; Ugolini, 2012).

Sweden is the exception that maps the frontier of feasible inclusiveness in the past. The Riksbank's remunerated accounts had no minimum amount; often account holders deposited as little as 100 kronor (corresponding to ca. 590 EUR as of October 2026), and

[22]The postal giro system was introduced in 1906, briefly before the Swiss National Bank started its operations in 1907.

the depositor base ranged from industrial companies, school boards, and businessmen to workers, janitors, maids, and postmen – alongside members of the royal family (Brisman, 1931). Two features made this possible: remuneration (which gave small savers a reason to come) and, importantly, an existing branch network. Even so, the business grew loss-making and was progressively restricted (Section 5.3). The German hyperinflation provides a revealing case of "accidental" inclusion. When commercial banks shed small customers to cut costs in 1922–24 (Deutsche Bank[23] alone cut its customer base from 804,000 to 281,000), small businesses and individuals flocked to the Reichsbank. The central bank promptly described the influx internally as an "undesired immigration," raised minimum balances, deferred new applications by six months, and forbade branches to advertise for deposit clients.[24]

Taken together, our evidence yields a clear distinction between *formal* and *effective* access. Clienteles emerged through (self-)selection induced by design: balance requirements, non-remuneration, fees, branch networks, and discretionary screening. Effective access depends on the cost of the access technology for the marginal user. Historically, that cost – idle balances, fees, distance – seems to have excluded precisely the populations that modern financial-inclusion arguments target.

5 Financial and macroeconomic implications

5.1 Cover ratios and monetary policy during the classical Gold Standard era

What "monetary policy" meant for a nineteenth-century note-issuing bank differed fundamentally from its modern counterpart. Countercyclical stabilization of output and inflation belongs mainly to the post-Great-Depression era. Instead, central banks operated under a convertibility mandate: their notes were redeemable in silver or, increasingly after the 1870s, gold at a fixed rate. Price stability, where it obtained, was a by-product

[23] See Annual Reports of Deutsche Bank, 1922–1924.

[24] Bundesbankarchiv, B332/8727 (secret directive of April 1923) (Reichsbank, 1925, p. 53). By 1925, 59% of private accounts held balances below 500 Reichsmark.

of the metallic peg (Flandreau, 2008; Ugolini, 2016). The chief task of the central bank was to defend, restore, or achieve convertibility.

Retail deposits entered central banks' historical mandates through one specific device: the cover rule, i.e., the regulatory requirement that made the convertibility mandate operational by tying note issuance to metallic reserves (proportionally, or through a fixed fiduciary allowance).[25] The cover ratio functioned as a real policy constraint because it implied a hard limit on the central banks' liquidity-providing lending operations. Cover rules were binding in a very concrete sense. First, they were written into statutes and charters, and violating them endangered the central bank's note-issue privilege – the foundation of its business. Second, cover ratios could also be easily observed. Most statutes coupled coverage with a disclosure requirement, requiring the bank to publish all relevant items on the balance sheet at monthly or weekly frequency. Importantly, the binding nature of the constraint did not depend on actual convertibility. For example, in Austria-Hungary, the florin had been inconvertible since 1866 and it was precisely for that reason that adherence to the statutory cover requirement was deemed of utmost importance. The rule remained "the only formal safeguard in place for the value of the Austrian currency" (Jobst and Rieder, 2016) – the anchor of public confidence in the value-storing capacity of banknotes.

Our sample of central banks divides into two configurations. In the first configuration – Belgium, the Netherlands and Switzerland – deposits sat inside the covered aggregate. The National Bank of Belgium required 33% of all sight liabilities to be covered by bullion since its creation. De Nederlandsche Bank had to hold metal against 40% of *all* sight liabilities, notes and current accounts alike. The first Swiss National Bank Act went furthest on paper. Art. 21 required *full* coverage of all short-term liabilities, including current accounts – but in high-quality assets including bills of exchange: a liquidity rule rather than a metallic reserve requirement. Between 1909 and 1911, the Swiss National Bank imposed a ten-day notice period on the withdrawal of minimum balances on deposit

[25] While cover ratios have a long tradition, earlier banks of issue did not always have a clearly stipulated rule. For example, Article 7 in the Royal Bank of Prussia's Revised Edict of 1766 states merely that banknotes should be issued "in a certain, non-harmful proportion" (Niebuhr, 1854, p. 204).

accounts, precisely so that these balances would no longer count as short-term liabilities subject to the statutory cover requirement (Swiss National Bank, 1932, p. 107; p. 317). The Swiss cover requirement was ultimately abolished in 1911, after which the Swiss National Bank maintained only an implicit, prudential cover practice.[26]

In the second configuration – the United Kingdom, France, Germany, Austria-Hungary, Spain, and Sweden – deposits lay outside the liabilities that needed to be covered. The English Bank Charter Act of 1844 simply did not mention deposit balances; the Banque de France's statutes of 1800 referred to the metallic reserve only in connection with the note issue, not with current accounts (Banque de France, 1880);[27] German law imposed no cover on deposits; the Austro-Hungarian Bank's current account liabilities were never subject to the silver (after 1892, gold) cover; the Banco de España's current accounts were excluded from the statutory cap on note circulation; and the Riksbank's deposits faced no reserve requirement (Thomas, 2024; Martín-Aceña et al., 2012).[28]

This asymmetric treatment of banknotes and current account deposits was the legislative legacy of the mid-century controversy between the Currency School – which held that only notes were money and had to be regulated – and the Banking School, which pointed out that deposits were functionally equivalent means of payment. The Currency School's victory, embodied in Peel's Act of 1844 and its continental imitations, wrote notes-only cover rules into most statutes (Fetter, 1965). Contemporaries were well aware of the resulting inconsistency: deposits were potentially unstable liabilities not subject to the regulatory discipline imposed on banknotes (Hübner, 1854, pp. 61–62).

To be sure, even where deposits were exempt, central banks were not entirely freed from the cover constraint. Account balances could be withdrawn in banknotes on demand because of the convertibility of deposits into banknotes at par. For this reason, current

[26] Nationalbankgesetz of 1905, Art. 20 (40% metallic cover on notes) and Art. 21 (Swiss National Bank, 1932, p. 317).

[27] France later moved to the first configuration. Art. 4 of the monetary law of 25 June 1928 required the Banque de France to hold gold equal to at least 35% of the sum of notes in circulation and current account balances.

[28] Despite their exclusion from the cover requirement, retail deposits in other countries were also subject to a liquidity requirement much like in Switzerland. For example, in Austria-Hungary, Article 84 of the 1878 central bank statutes reads: "The amount of banknotes and other sight liabilities that exceeds the metallic reserves has to be covered by bills discounted, advances on specie or foreign bills coupons that are due."

account deposits were effectively contingent notes in circulation, pre-committing scarce issuance headroom that the cover rule rationed. The Austro-Hungarian case shows this implicit constraint at work. Until 1888 the central bank operated under a strictly binding note ceiling – metallic reserves plus a fixed fiduciary issue of 200 million florins, on the model of the Bank of England – and hesitated to expand its current account business for fear that deposit withdrawals would push it into violation of its statutes. The statute renewal of 1888 kept the ceiling but made it soft: temporary transgressions became legal provided the central bank maintained a proportional metal cover of at least 40% and paid a compensatory tax on the excess issue – a rule-based flexibility known to the public *ex ante* (Jobst and Rieder, 2016). Only then, with the latent constraint relaxed, did the bank introduce inter-branch transfers and expand its deposit and giro business (Bubenik, 1888; Rauchberg, 1897). Hence, the formal exemption of deposits from cover requirements provided no real comfort while the note ceiling was not flexible: what mattered economically was not whether deposits were counted in the covered aggregate but that they were convertible claims on this aggregate.

Still, where only notes had to be covered, the measured cover ratio mechanically improved when the central bank retired covered liabilities by shifting them into current accounts. Crediting the current account with banknotes or specie would, other things equal, augment the gold cover of notes in circulation. Depositing banknotes would reduce the denominator of the cover ratio, whereas specie deposits would increase its numerator. Substituting account money for note circulation in the settlement of payments had the same effect at scale: it reduced the volume of metal cover a given quantity of transactions required.

Central banks understood and used this margin. The Reichsbank, stripped by the Bank Act of 1875 of unlimited note issuance and of interest-bearing deposits, pivoted into the liability the law left unconstrained – the unremunerated giro deposit – and acknowledged that the strategy was driven by the legal architecture rather than by demand. It promoted cashless settlement systematically, raising the share of giro settlements executed without cash from 72% in 1886 to 90% in 1914. The central bank explicitly stated that

"the advantages of combining banknote issuance with giro transactions are particularly evident in the fact that the cash flowing to the central banks through giro transactions reduces their unbacked banknote circulation" (Reichsbank, 1901, p. 51). Spain, where the peseta was not convertible to gold, provides the mirror image under stress: during the 1931 crisis, as hoarding drove up note circulation, the financial press repeatedly urged the public to deposit hoarded cash in Banco de España's current accounts precisely because this would reduce circulation, improve the "apparent" gold cover of the peseta and help stabilize the exchange rate (Jorge-Sotelo, 2026).

The consequence of this form of "regulatory arbitrage" was that the central banks in question benefited from more scope to expand their lending operations, with positive effects on their profitability. Because current account balances were unremunerated and, in terms of metal cover, free resources, they constituted costless funding that could be invested in the discount and Lombard portfolio. The Austro-Hungarian Bank expanded its current account business after 1888 precisely because it "allowed the expansion of interest-free liabilities that could be invested in the discount and lending business" (Bubenik, 1888; Rauchberg, 1897). The Bank of England's requirement that current accounts be "remunerative" (see Section 4.2) reflected the same calculus. In Sweden, the Riksbank invested the deposit inflows in its lending business. Interestingly, the Riksbank also held its deposit and lending rates deliberately below market in an attempt to pull down the general level of interest rates (Brisman, 1931). Deposit pricing subordinated to an interest-rate objective is the closest nineteenth-century counterpart to the modern idea of using the remuneration of central bank money as a policy instrument (Bordo and Levin, 2017).[29]

Where deposits were inside the covered aggregate, the arbitrage margin disappeared, but so, in practice, did the constraint – because sums deposited in current accounts were small in the respective countries. Dutch current account balances never exceeded 1% of notes in circulation before 1890 and stood below 10% of the balance sheet even at their 1914 peak. The 40% metal cover was binding through note issuance only. When the

[29] Apart from gauging the stance effects of bank disintermediation, modern analyses of the monetary policy implications of retail CBDC have *inter alia* focused on its potential to relax the effective lower bound on interest rates (Bordo and Levin, 2017; Rogoff, 2017).

crisis of 1914 stressed the system, the pressure came from the demand for gold, silver, and banknotes – the deposit inflow was an order of magnitude smaller than the increase in note issuance – and the government temporarily halved the cover ratio to 20% (de Jong, 1930; Jongman, 1960).

The Dutch case thus serves to make another general point: whether formal inclusion in the cover mattered depended on the quantitative importance of deposits relative to notes. That importance varied widely (see Figure A1 in Appendix A). At one end, retail deposits were substantial: at the Bank of England they averaged 26% of Banking Department liabilities between 1844 and 1914, with a ratio to notes in circulation averaging 0.53 (Antipa and Rieder, 2026); in Spain they reached about 30% of central bank liabilities in the 1880s–90s; in Sweden the remunerated deposit business peaked in 1879 at 22 million kronor, at the time around 20% of the Riksbank's balance sheet. At the other end, balances remained marginal – for example, in Austria-Hungary (where turnover, not the stock, was large), Germany and Switzerland (both with retail giro balances of the order of 3–9% of notes in circulation). Whether these stark differences were endogenous reactions to the legal specification of the cover requirement, or reflected design choices that made holding current accounts unattractive for end users – choices which may themselves have responded to the cover requirement – remains difficult to disentangle.

5.2 Flight-to-safety dynamics during crises

If retail deposits were not neutral for the conduct of monetary policy in normal times, crises made their potential financial and economic impact even more visible. History shows that central banks' current accounts regularly served as the depositary of last resort when the public's confidence in private alternatives was shaken.

Banking panics – crises centered on the solvency or liquidity of commercial banks – and political shocks often generated flights into central bank accounts. During the Overend-Gurney panic of May 1866, retail deposits at the Bank of England rose by some £5 million against a pre-crisis stock of roughly £9 million, and the Bank's share of the London deposit market jumped from 11% to 19% (Antipa and Rieder, 2026). The Riksbank's retail

business surged during the crisis of 1878–79: the number of accounts at the head office more than tripled between 1876 and 1880, and 1879 marked the all-time peak of its deposit business (Brisman, 1931). Private balances at the Prussian Bank rose by more than four-fifths during the revolutionary crisis of 1848–49 (Königlich-Preußische Hauptbank, 1850, p. 7). Private deposits at the Banque de France likewise show discernible inflows in 1848 and during the crisis of 1889 (Baubeau, 2018; Hautcoeur et al., 2014).[30] In the Netherlands, the panic of August 1914 – triggered by the outbreak of World War I – drove current account balances up by 30 million guilders within weeks, lifted the share of current accounts in the balance sheet from about 1% to 10%, and quintupled the number of household and firm accounts by year-end (Mulder, 2026). During the same period, four of the five largest commercial banks lost $f.$ 90 million deposits in the first days, equal to a quarter of their balance sheets (de Jong, 1930). British entry into World War I also led to an explosion of retail deposits at the Bank of England (Antipa and Rieder, 2026). Similarly, in Austria, current account balances at the central bank shot up when the Creditanstalt's losses became public in May 1931 (Oesterreichische Nationalbank, 1932). In France, depositors moved funds from commercial banks to savings institutions and to the Banque de France during the banking crises of 1930–31, when the central bank's current accounts of individuals increased strongly (Baubeau et al., 2021). In Spain, after the proclamation of the Second Republic in April 1931, banks lost 1,216 million pesetas of deposits, while Banco de España saw an inflow of 273 million pesetas in retail deposits, a roughly 40% average increase across cities. This flight-to-safety response scaled with the severity of the bank run at the city level (Jorge-Sotelo, 2026).

One comparative finding of our country studies is that not all historical episodes of distress triggered broad recourse to the depositary of last resort. For example, several crises that called into question the continuity of the central bank itself – or of the sovereign behind it – produced no such flight. Safe-haven demand for central bank deposits required the central bank to be both more solid than the private alternatives and institutionally secure in its own right. The Royal Bank of Prussia *de facto* defaulted during the Napoleonic

[30]The French series of private deposits includes banks' balances and has gaps around 1870–71. Its movements should therefore be interpreted with caution (Baubeau, 2018).

invasion of 1806. Its private deposits fell by nearly a quarter and never regained their pre-war level (Niebuhr, 1854, pp. 227–228). Deposits drained from the Riksbank on news of invading troops in 1700, and the run of 1709 – triggered by the defeat at Poltava – forced the suspension of convertibility of its lending-bank deposits (Wetterberg, 2010; Fregert, 2018).[31] During the Austro-Hungarian banking panic of November 1912, which unfolded under the acute threat of a Russian invasion via Galicia during the Balkan crisis, current account balances stayed within their ordinary seasonal pattern: deposits did not flee the central bank, but no safe-haven flow into central bank deposits is discernible (Jobst and Rieder, 2023). The French record is similarly mixed. Private deposits at the Banque de France show no significant movements during the crises of 1857 (Juglar, 1862) and 1882 (White, 2007), and they recorded a sizable outflow in 1873, in the aftermath of the military defeat of 1870–71 and the ensuing political and social crisis (Baubeau, 2018).

When the loss of confidence in the commercial banking system hit, the central banks' retail deposit customers had the choice between a flight into their current accounts and a flight into cash. These two safe-haven assets were liabilities of the same institution and convertible at par. Differential credit risk therefore cannot explain why the public may have preferred one over the other. And yet, we observe a consistent pattern: flights into cash dwarfed those into current accounts during panic moments. For example, during the English panic in 1866, the public exhibited a strong preference for cash relative to deposits: borrowers from the Bank of England – virtually all of whom had drawing accounts – became 10 to 30 percentage points less likely to have loan proceeds credited to their accounts after the panic broke, taking banknotes instead (Antipa and Rieder, 2026). The relative preference for banknotes is also visible in aggregate flows elsewhere. The balance sheet of De Nederlandsche Bank increased by 47% in the month after the outbreak of World War I. 80% of this increase can be attributed to banknote issuance, while only 18% can be explained by inflows into current accounts.[32] In both major Swiss crises of the period under observation, hoarding of banknotes and metal dominated:

[31] Established in 1668 as the only bank in the country, the Riksbank accepted both non-remunerated deposits and remunerated retail deposits from its inception (Brisman, 1918; Fregert, 2018).

[32] Nationaal Archief, Den Haag, De Nederlandsche Bank, 2.25.68 - 100103: DNB daily balance sheets, 1914–1915.

during the First World War, notes in circulation more than tripled while giro balances rose modestly; in the Great Depression, the share of 1,000-franc notes in circulation nearly doubled within a year. Central bank current account balances did rise more than fivefold between 1929 and 1932, but the increase was concentrated among banks and large firms (Swiss National Bank, 1932). Even during the Austro-Hungarian panic of 1912, when no increase in central bank current accounts occurred, the demand for banknotes rose sharply.

One possible explanation for this preference for cash over central bank accounts is rooted in the different services the two instruments provide. Banknotes are a bearer instrument: immediately usable, transferable without infrastructure, and robust to the disruption of the banking and the account-keeping apparatus. A deposit balance, by contrast, is only as accessible as the branch and the transfer system that administers it. Where the anticipated disruption extends to the payment infrastructure itself – as in an expected direct war involvement or a full-scale invasion – the premium on the bearer instrument becomes even greater, and hoarding notes (or metal) dominates crediting a current account.

Despite this apparent dominance of cash during crises, the coexistence of these two safe havens can be explained by several non-mutually-exclusive factors. First of all, the central bank current accounts were mostly held by a select group of wealthy firms and households, who likely diversified their liquid assets in times of crisis. Second, central bank current accounts could still be used as a means of payment using the central bank branch network during crises, whereas interbank transfers between local commercial banks might incur higher uncertainty and costs. Third, preferences for keeping cash at home – and the costs associated with it – might have differed between households, with some facing higher costs of holding physical cash (e.g., risk of theft and limited storage capacity) and others perceiving the central bank vault as a safer place to store their money.

The observed preference for cash also allows us to connect the crisis record back to the cover rules discussed in the previous subsection. Panics produced internal drains – transfers of commercial bank deposits and less liquid assets into central bank current

accounts, banknotes and specie – that frequently brought central banks close to violating their cover requirements. In principle, a crisis-driven shift of the public from notes into central bank deposits could have relaxed the constraint. In practice, however, the pronounced preference for cash meant that liquidity crises tightened rather than loosened the constraint, and any deposits gained remained convertible claims on the metallic reserve under pressure. The composition channel that may have helped in normal times appears to have offered little relief precisely when it would have been most valuable. Central banks or their governments were repeatedly forced to relax or suspend cover requirements during panics. The English Bank Charter Act was suspended in 1847, 1857, and 1866, the Austrian cover requirement in 1873, and the Dutch metallic cover was halved in 1914 (Thomas, 2024; Jobst and Rieder, 2016; Jongman, 1960).

5.3 Competition and disintermediation

Two key questions in the current debates surrounding CBDC are whether the introduction or availability of CBDC will lead to a systematic and lasting disintermediation of the commercial banking sector, and if so, under which conditions this disintermediation will have negative macroeconomic consequences. In our historical context, a natural starting point for approaching these questions is to ask whether crisis inflows into central bank current accounts proved persistent.

The historical record suggests that "ratchet effects" were not automatic consequences of large crisis inflows. In our sample, we observe both transitory spikes and durable level shifts. For example, following the 1848 turmoil, Prussian Bank balances were back near their pre-crisis level by 1851. The spikes around the Baring crisis of 1890, the British crisis of 1931, and the Austrian Creditanstalt episode of 1931 also reversed quickly. By contrast, non-financial firm accounts newly opened at the Bank of England in 1866 exhibited a durable ratchet. Dutch balances likewise remained elevated throughout World War I even after the acute flight reversed, yet the aggregate numbers cannot disentangle bank from retail deposit balances (Mulder, 2026). Spanish central bank deposits after 1931 stayed 20–40% above their pre-crisis level for years (Jorge-Sotelo, 2026). In France, the

inflow of 1848 appears to have shifted the level of private deposits at the Banque de France quasi-permanently, although the underlying series is noisy and includes banks' balances (Baubeau, 2018). For the later nineteenth century, by contrast, Théry (1895, p. 19) reports that large commercial firms moved deposits to the Banque de France only in times of commercial and financial crisis, and suggests that these inflows were short-lived because more remunerative alternatives were available.

Based on the English case, Antipa and Rieder (2026) discuss the conditions that could turn a spike into a ratchet: a low post-crisis interest rate environment that compresses the opportunity cost of unremunerated central bank balances; positive network externalities, as the panic coordinates many previously dispersed customers onto the same institution and raises the value of same-institution transfers; and the composition of the new clientele, with financially conservative account holders retaining their accounts. The Spanish evidence adds a fourth element: the destruction of local private alternatives and scarring effects. After 1931, the persistence of higher balances with Banco de España was concentrated in Spanish cities where locally large banks had actually failed during the crisis. Hence, ratchet effects appear to arise when design features or crisis damage durably change the relative attractiveness of the central bank account.

The Dutch case offers a complementary insight: even in the absence of ratchet effects, commercial banks themselves chose to diversify their funding following flights into central bank current accounts. Mulder (2026) shows that commercial banks that had more geographical overlap with the central bank branches increased their deposit rates by more than their less exposed peers during the crisis months. As the crisis subsided, deposits largely returned to all banks. In the following years, however, more exposed banks relied less on deposit funding and grew slower. Because this happened after deposits had returned and rate differentials had disappeared, Mulder (2026) interprets it as a choice by the banks rather than a lasting loss of depositors: the crisis led exposed banks to reassess deposit funding as less stable.

Whether persistent disintermediation resulting from a flight into central bank deposits has macroeconomic consequences depends on both how the central bank reacts to inflows

and how commercial banks respond to outflows. In their framework, Brunnermeier and Niepelt (2019) argue that a swap of private inside money (bank deposits) for public outside money (central bank deposits or CBDC) leaves the equilibrium allocation unchanged provided the central bank passes the funds it receives back to the banking sector. For strict equivalence of private and public money, central bank funding needs to be provided on the same terms as the deposits being replaced, so that banks' balance sheet size and funding conditions are preserved and only the composition of their liabilities changes. Under these conditions, fears of disintermediation are misplaced: the migration of deposits to the central bank is a neutral reshuffling, and the central bank effectively intermediates between the public and the banks. Alternatively, commercial banks can respond to deposit outflows by offering more attractive contracts (Ahnert et al., 2025; Mulder, 2026) or by diversifying into liabilities other than central bank borrowing. The cost of banks' funding will likely increase as a result, as bank deposits are a cheap source of funding. This can ultimately lead to disintermediation if banks pass through the higher funding costs into loan rates or reduce their loan supply. For example, Mulder (2026) provides suggestive evidence that Dutch banks more exposed to central bank competition for deposits expanded their lending less in the years after the 1914 panic. Slower lending growth was particularly pronounced in illiquid loans of exposed banks, consistent with the notion that these banks treated deposit funding as less stable.

Our historical case studies provide several insights related to the ongoing discussions about the macroeconomic consequences of disintermediation. First, at the most basic level, we find evidence that central banks did not particularly welcome crisis-driven inflows. For example, the Prussian Bank's management regarded the 1848–49 capital flight into its accounts as an aggravating factor that deepened the crisis. The Riksbank repeatedly stopped accepting new deposits when it could not invest them and the Reichsbank at times met a strong influx with higher entry barriers. The President of De Nederlandsche Bank even spoke about the public that gave "the Bank a credit it did not ask for" (Vissering, 1917, p. 168).

Second, the historical actors in our study did not behave as though they expected the

equivalence theorem to hold. Competition for deposits was treated, almost universally, as a zero-sum contest – likely because many central banks evolved out of private banks or remained in private ownership throughout our sample period. Several central banks deliberately abstained from the retail market for this reason. The National Bank of Belgium, founded in 1850 with the note-issue monopoly, was designed not to take retail deposits. It was barred from paying interest on its liabilities, restricted to accounts serving the discount business, and expected to leave savings collection to the commercial banks and, after 1865, the state savings bank (Chlepner, 1926; Kauch, 1950; Ugolini, 2012). De Nederlandsche Bank was prohibited from taking deposits outright at its foundation in 1814. Even after the prohibition was lifted in 1838, its leadership disavowed any intention to compete. Deposit-taking, in the words of President Mees, was the business of banks and *kassiers*, not of the central bank. According to Vanthoor (2005), this mantra led De Nederlandsche Bank to keep the terms of its current accounts deliberately unattractive. The Banque de France did not compete for deposits either. Its current accounts remained unremunerated, and while the number of deposit accounts at the four largest commercial banks almost tripled during the last decade of the nineteenth century, the number of accounts held at the Banque de France declined over the same period (Théry, 1895, pp. 50–52).

Where the central bank itself came to hold a dominant position on the retail deposit market, growing commercial banks were perceived as a threat. The Prussian Bank's deposit dominance had been partly manufactured by regulation: between 1848 and 1857 it was the only note-issuing bank permitted to accept interest-bearing deposits, and by 1864 it held 85% of all deposits at Prussian note-issuing banks (Roepell, 1868, p. 7). Its successor, the Reichsbank, inherited this position. As a corollary, when the German universal banks expanded into the provinces after 1895 – Deutsche Bank's deposits grew from 56.3 million Mark in 1895 to 732.6 million Mark in 1913,[33] with total accounts moving from under 1,000 in 1876 to over 50,000 by 1900 – the Reichsbank perceived the growth of private deposit banking as a direct assault on its "free" funding base and

[33]See Annual Reports of Deutsche Bank, 1895 and 1913.

contemplated reintroducing interest-bearing deposits. In short, it responded much as a threatened commercial competitor would (Bopp, 1953, p. 78).

The political economy of the competition for retail customers had a recurring focal point: the granting of the note-issue monopoly was typically the moment at which the central bank's retail role was settled. These negotiations went in opposite directions in different countries. The same instrument – the conferral of the note monopoly – was used both to evict the central bank from the retail deposit market and to draw it in. In Spain, the decade following the note monopoly of 1874 was associated with an enlargement of the Banco de España's retail footprint through the expansion of its branch network into underbanked regions, causing a large increase in central bank retail deposits in the 1880s (see Figure 1). After initial hesitation due to unprofitability and lack of local capital to fund new branches, the latter began performing a leading role in the local retail banking market – many provincial capitals had no banking presence at all before the central bank arrived there during the 1880s (Castañeda, 2001). By contrast, in Sweden, the Riksbank Act of 1897 conferred the note monopoly and simultaneously obliged the Riksbank to wind down its interest-bearing retail deposits. Interest payments on retail deposits ceased in 1899 and the sum stored in current accounts fell by 94% within a year. The statutory eviction ratified a retreat that had begun internally a decade earlier. As the Riksbank's commercial lending was restricted in the late 1880s on its way to becoming a "real" central bank, the deposit business had lost its *raison d'être*, and an 1888 motion to stop paying interest had only narrowly failed (Brisman, 1931; Wetterberg, 2010). The Swiss example also suggests that coexistence could be cooperative rather than competitive as central banks' roles evolved. The Swiss National Bank and the Swiss postal giro settled into serving different clienteles – wholesale and retail, respectively – and integrated their systems with free mutual transfers, the central bank acting as the apex of a two-tier payment system rather than as a retail competitor (Swiss National Bank, 1957).

Third, historically, even when the central bank was willing to channel liquidity to banks in a crisis, the equivalence benchmark may not have held for a technical reason. The neutrality result in Brunnermeier and Niepelt (2019) requires the pass-back to take

the form of neutral refunding. Banks that lose deposits obtain a replacement liability on unchanged terms, leaving their balance sheet intact.[34] In the historical context, however, the dominant channel of central bank liquidity provision was not collateralized lending but discounting – the outright purchase of bills of exchange. While banks would need to possess (high-quality) eligible assets to access the central bank's lending facilities, the distinction likely mattered: a commercial bank that paid out fleeing deposits with the proceeds of rediscounted bills had to sell assets to the central bank, and its balance sheet contracted on both sides (Figure 3, Panel A). In contrast, a bank that paid out depositors with the proceeds of an advance took on a new liability, preserving its balance sheet size (Figure 3, Panel B). Under a discounting-based pass-through, the migration of deposits to the central bank was therefore not funding-neutral for the banking sector: it deleveraged, and its intermediation capacity contracted, with direct consequences for the supply of credit.[35]

Figure 3: **Paying out depositors: discounting vs. collateralized borrowing**

Panel A. Discounting

Assets	Liabilities
−100 Bills of exchange +100 Cash / Deposits at central bank	
−100 Cash / Deposits at central bank	−100 Deposits

Panel B. Advances (Lombard loans)

Assets	Liabilities
+100 Cash / Deposits at central bank	+100 Central bank advance
−100 Cash / Deposits at central bank	−100 Deposits

Note: This figure shows the two options historically available to a commercial bank to obtain liquidity from the central bank when faced with a deposit funding shock during a bank run.

For example, in 1866, the Bank of England acted as a full-fledged lender of last resort,

[34] Grodecka-Messi and Zhang (2025) study a related issue linked to the loss of note issuance by private banks, examining the role of central bank liquidity support in this context.

[35] These mechanics applied to financial institutions that operated as intermediaries, i.e. institutions that mainly *discounted* bills for their customers before re-discounting them at the central bank. They were likely less relevant for discount or merchant banking houses that also *originated* bills on a large scale.

but primarily through the discount window – so that the very act of accommodating the crisis translated the deposit flight into a contraction of commercial bank balance sheets (Antipa and Rieder, 2026). Although the central bank could have directly recycled funds to the economy by purchasing more securities from the non-bank private sector than before the crisis, it did not do so. Similarly, during the Spanish bank run in 1931, the banking system lost more than 20% of its deposits. This funding shock caused affected banks to restrict lending to individuals and firms, leading to a credit crunch. While the Banco de España provided advances and discounts to the banking sector (Jorge-Sotelo, 2020), it did not make systematic use of deposit inflows to provide additional loans to non-financial firms or individuals to replace banks' lost intermediation capacity. The central bank had no mandate or mechanism to redistribute this sudden increase in liquidity through its branch network and lacked the local screening and monitoring capacity of banks whose deposits it had absorbed (Jorge-Sotelo, 2026). The French banking crises of 1930–31 followed a similar pattern. According to Baubeau et al. (2021), the funds that moved from banks to savings institutions and the Banque de France were not recycled into credit: gold reserves were the only asset item on the central bank's balance sheet that grew massively, its direct lending to non-financial firms remained negligible, and no other financial institution replaced the banks as lenders. The result was a severe and persistent credit crunch.

6 Conclusion

This paper assembles new comparative evidence on the closest historical precedent of retail CBDC: the deposit accounts that European central banks offered to individuals and firms other than banks until the early twentieth century. We specifically focus on a transitional period in the history of financial intermediation, when central bank retail deposits existed alongside commercial bank deposits.

Our findings can be summarized as follows. Central banks offered retail deposits to build national payment infrastructure and to complement their lending operations. In normal times, customers used accounts mainly for liquidity management and payment

purposes. Access that was broad in law was narrow in fact, because pricing, balance requirements, discretion, and geography reserved effective use to firms and wealthy households. Demand for the accounts existed even on unattractive terms, but it rested on the services bundled with the account, and balances were minimized accordingly. Retail deposits were not neutral for monetary policy: often exempt from cover requirements yet convertible into notes on demand, they served as an instrument for stretching the cover, but also brought risks as contingent claims on constrained note issuance. Finally, central bank deposits acted as a conditional safe haven during crises. Inflows did not always persist, and their impact on the banking system depended on the liquidity-providing operations the central bank used to recycle the funds.

The historical environment differed from today's along several important dimensions: monetary policy meant defending convertibility, deposit insurance did not exist, payments were note-based, and central banks were often still in private hands. Yet, the historical evidence on the gap between formal and effective access, the dependence of demand on bundled services and remuneration, the interaction of retail central bank money with the operational framework of monetary policy, the conditionality of safe-haven flows and ratchet effects, and the decisive role of the recycling channel for disintermediation provide valuable insights for current discussions. Ultimately, the relevance of the historical precedent we have analyzed in this paper will also depend on the design features chosen for CBDC. For example, very restrictive holding limits or unattractive remuneration may cause the take-up of future CBDC to mimic the modest recourse to central bank retail deposits in some European countries in the past – and vice versa.

Our findings suggest some avenues for future research. First, more work on data generation, including for central banks we do not cover in this paper, would be highly welcome. Additional harmonized cross-country series on account numbers, balances, and turnover would allow the record assembled here to be put to quantitative uses – for example, to test how specific design features affected take-up, or to substantiate the payment-rails interpretation with turnover-to-balance ratios. New efforts to separate retail deposits from bank deposits held at the central bank would be equally valuable.

Second, additional micro-level studies of the kind available for the Netherlands, Spain and the United Kingdom would allow for a more detailed and deeper look at the remaining cases of our sample and beyond. Third, the historical political economy of central banks' entry into – and eventual retreat from – the retail deposit business deserves a more systematic treatment.

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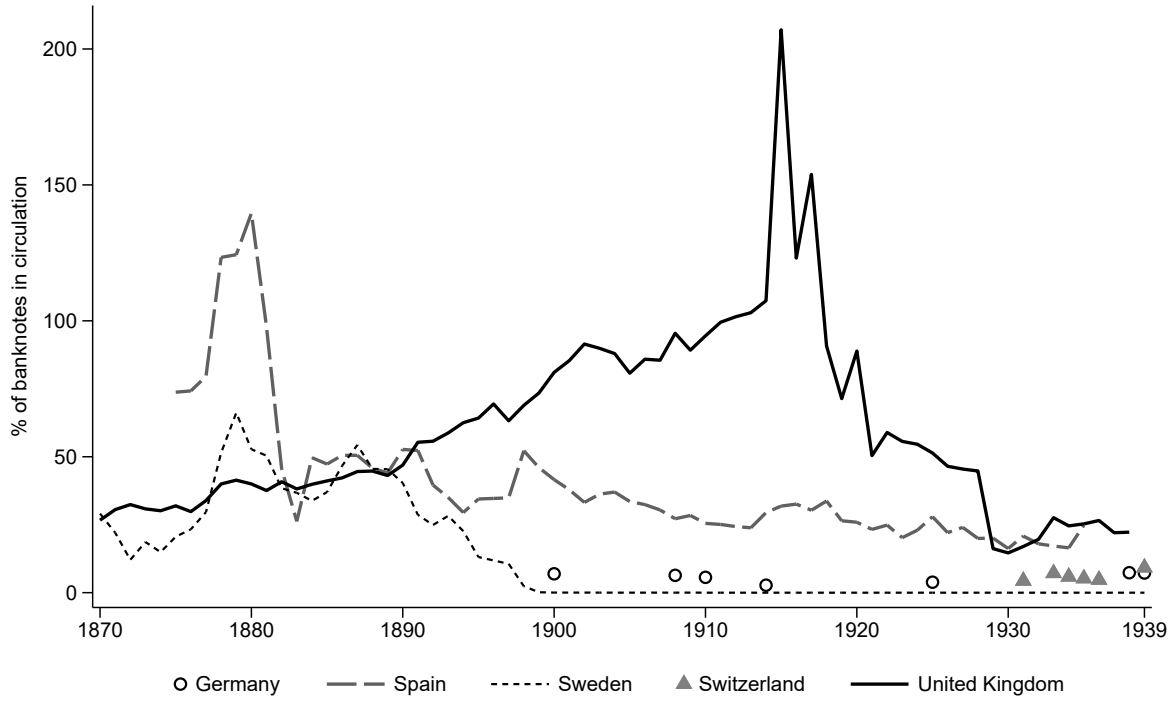
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APPENDIX

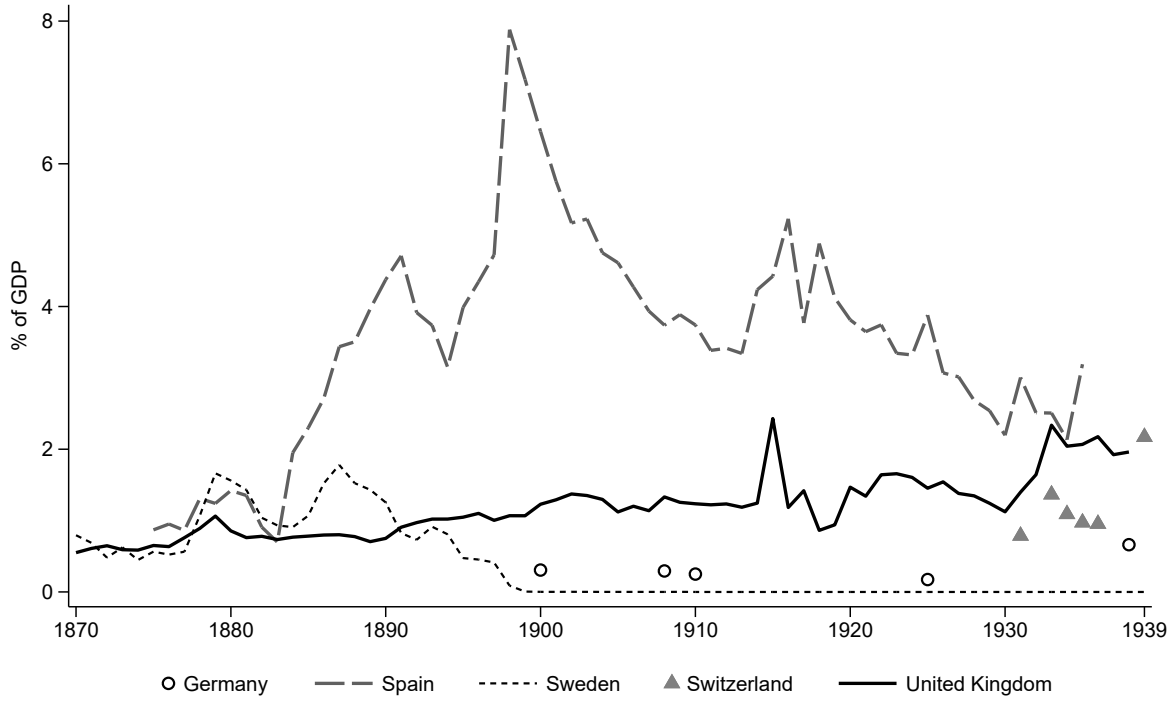
A Additional figures

Figure A1: **Central bank retail deposits as % of banknotes in circulation, 1870–1939**



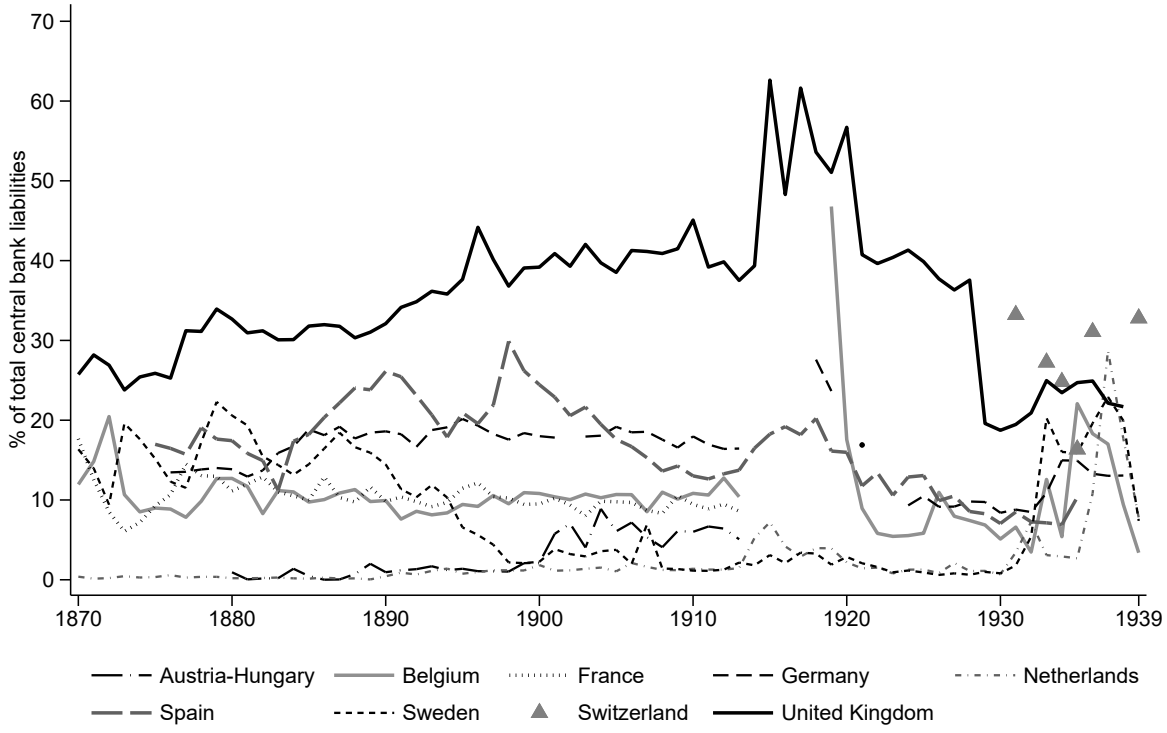
Note: This figure shows central bank retail deposits in five European countries as % of the total notes in circulation. We define retail deposits as deposit accounts held by individuals and private sector firms other than banks. Sources: see Appendix B.

Figure A2: **Central bank retail deposits as % of nominal GDP, 1870–1939**



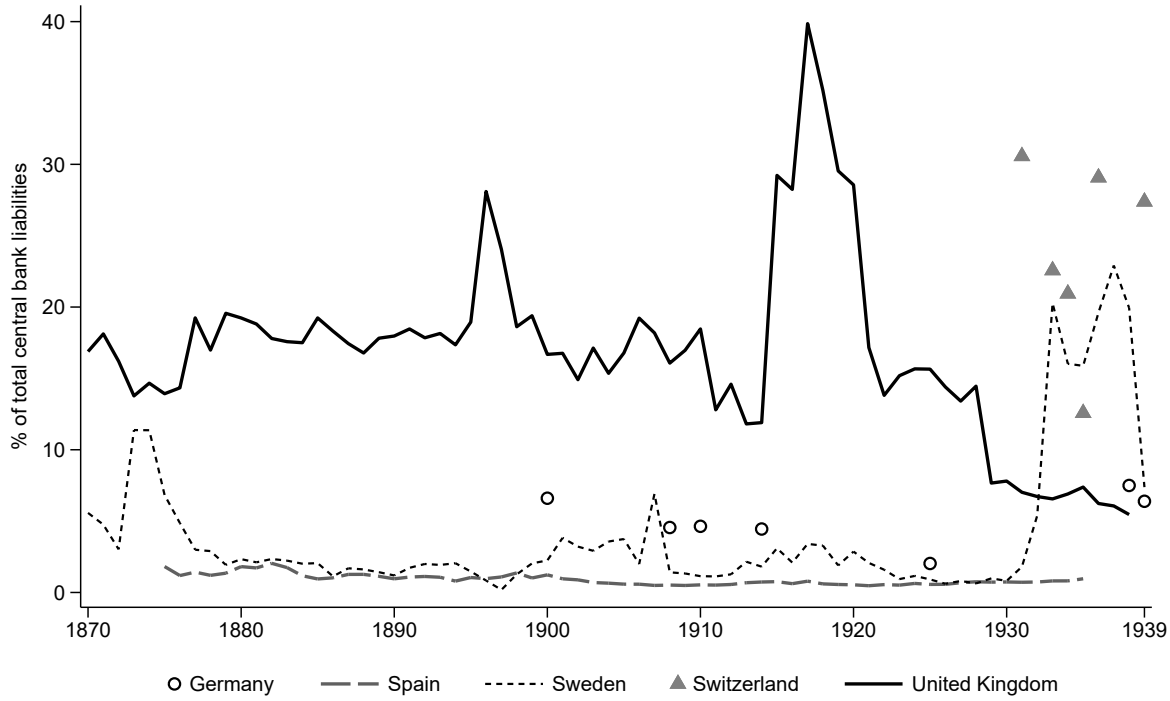
Note: This figure shows central bank retail deposits in five European countries as a share of nominal GDP (in percent). We define retail deposits as deposit accounts held by individuals and private sector firms other than banks. Sources: see Appendix B.

Figure A3: **Total private sector deposits at the central bank as % of total central bank liabilities, 1870–1939**



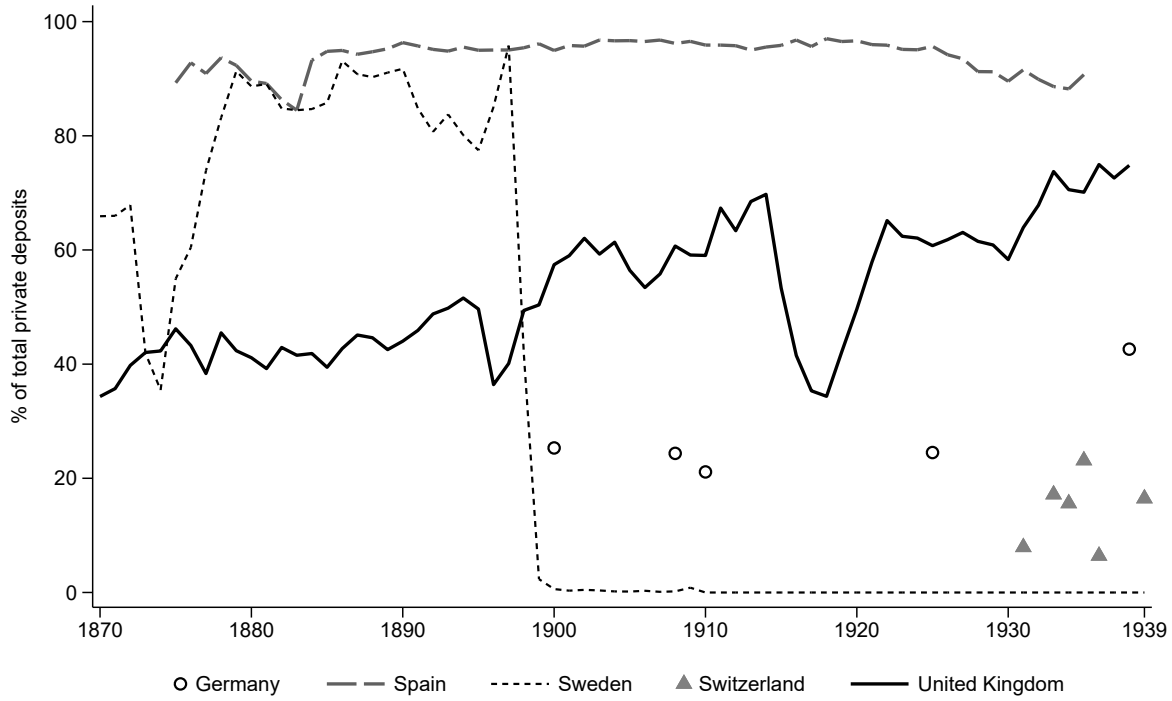
Note: This figure shows total private sector deposits at the central bank (retail deposits plus deposits by banks and other financial intermediaries) in nine European countries as a share of total central bank liabilities (in percent). We define retail deposits as deposit accounts held by individuals and private sector firms other than banks. Sources: see Appendix B.

Figure A4: **Commercial bank deposits at the central bank as % of total central bank liabilities, 1870–1939**



Note: This figure shows deposits held by banks and other financial intermediaries at the central bank in five European countries as a share of total central bank liabilities (in percent). Sources: see Appendix B.

Figure A5: **Retail deposits at the central bank as % of total private sector deposits at the central bank, 1870–1939**



Note: This figure shows central bank retail deposits in five European countries as a share of total private sector deposits at the central bank (in percent). We define retail deposits as deposit accounts held by individuals and private sector firms other than banks. Total private sector deposits encompass both retail deposits and deposits by banks and other financial intermediaries. Sources: see Appendix B.

B Archival and secondary sources

B.1 Austria-Hungary

- Central bank deposits:
  - Source/Archive: Annual reports
    - 1870–1878: General-Versammlungen der priv. österreichischen National-Bank.
    - 1879–1884: Jahressitzung der Generalversammlung der Österreichisch-Ungarischen Bank.
    - 1885–1914: Regelmäßige Jahressitzung der Generalversammlung der Österreichisch-Ungarischen Bank.

B.2 Belgium

- Central bank deposits:
  - Source/Archive: National Bank of Belgium (1950).

B.3 France

- Central bank deposits: Baubeau (2018). The available series of private deposits does not separate banks from other account holders.
- Rules governing current accounts: Banque de France (1830, 1880); Théry (1895).

B.4 Germany

- Central bank deposits: Deutsche Bundesbank (1976).
  - Source/Archive:
    - Bundesbankarchiv, B332/8727 Schriftwechsel und Verfügungen an die Reichsbankstellen.

        - ∗ Bundesbankarchiv, B332/22 Geheime Verfügungen des Reichsbankdirektoriums und Antwortschreiben bzw. Berichte der Reichsbankstelle Emden.

- Nominal GDP (net national product): Burhop and Wolff (2005); Ritschl and Spoerer (1997); Statistisches Bundesamt (1961); Hoffmann (1965).

B.5 Netherlands

- Central bank deposits: The available annual series of private deposits does not separate banks from other account holders.
    - DNB archive:
        - ∗ 1870–1913: 2.25.68 (from 101380 to 101381)
        - ∗ 1914–1920: 2.25.68 (101382)
        - ∗ 1921–1939: 2.25.68 (from 101382 to 101383)

- Nominal GDP: CBS 200 jaar statistiek (1870–1913 and 1921–1929); CBS Berekeningen over het nationale inkomen van Nederland voor de periode 1900-1920 (1914-1920).

B.6 Spain

Data for Spain is collected from the Banco de España Historical Archive (AHBE). Banco de España (BdE) annual reports (*Memorias*) provide total current accounts held by the government and by private agents ("*cuentas corrientes de particulares*"). Thus, the BdE did not report banks' current accounts separately from accounts for firms other than banks and individuals. However, the secondary sources detailed below allow to subtract the amounts deposited by banks, saving banks and other financial institutions, from the aggregate accounts provided by the BdE, thus resulting in the amount of retail deposits shown in Figures 1, A1, A2 and A5. Overall, as Figure A5 documents, the vast majority of "*cuentas corrientes de particulares*" were held by individuals and firms, as banks used other facilities ("*cuentas corrientes de crédito*") to borrow from and park reserves at the BdE (Jorge-Sotelo, 2026).

- Central bank deposits and central bank balance sheet:
  - Banco de España historical archive (AHBE):
    - For "*cuentas corrientes de particulares*" and total liabilities *Memorias del Banco de España*, 1875–1935 (Signature: 3/5A/4/1).
    - For regulation and operationalization of accounts:
      - *Instrucciones para el Régimen de las Sucursales del Banco de España*, various years (Signature: L.164).
      - *Estatutos y Reglamento del Banco de España*, various years (Signatures: L. 102, 153, 154, 216, 258, 281, 283, 324, 352, 354, 504).
  - To be able to separate Banco de España current accounts held by individuals and firms other than banks (retail deposits) from those held by financial intermediaries we obtain the share of BdE deposits that corresponds to banks, savings banks and other financial intermediaries, using the figures on "*cuentas corrientes BdE*" provided by Martín-Aceña (1985). These are available for the period 1900–1935 and provide total current accounts held by these financial intermediaries at the BdE. These are reported separately as part of the intermediaries' cash reserves, so they reflect the exact end-of-year balance of financial intermediaries held at their current accounts with the BdE. For the period 1875–1899, the size of BdE retail deposits has to be estimated. This is done by extrapolating backwards the share of BdE current accounts held by banks, savings banks and other financial intermediaries over their total reserves using data published in Tedde and Tortella (1970), Tedde (1970) and Tortella (1970).
- Nominal GDP, 1870–1935: Prados de la Escosura (2003, Cuadro A.7.).

B.7 Sweden

Available Sveriges Riksbank sources report deposits by type of account, not by account holder. At times, however, certain types of accounts were offered to a specific clientele

only. By combining the aggregate data with historical narratives and the accounts of state agencies, we approximate retail and bank deposit holdings for years in which more detailed reporting is not available, as described below.

- Central bank deposits:
  - Long-run data on deposits at the exchange and lending (loan) bank are available in the appendix to Fregert (2014).
  - Data on non-interest-bearing and interest-bearing accounts for 1864–1924 come from Sveriges Riksbank (1931).
  - Interest-bearing retail deposits are the sum of current accounts (*upp- och avskrivningsräkning*) and time deposits (*depositionsräkning*) in Sveriges Riksbank (1931), minus the interest-bearing deposits of the National Debt Office (from Riksdagens revisorer, 1870a) and of the postal savings bank and other state agencies (from Riksdagens revisorer, 1870b). In 1873 we also deduct deposits attributable to banks, inferred from the increase in non-interest-bearing deposits between 1873 and 1874. For 1870–1872 bank and retail deposits cannot be separated; from 1874 banks could no longer hold interest-bearing accounts at the Riksbank, so their funds moved to non-interest-bearing accounts.
  - Non-interest-bearing accounts of banks and other private depositors:
    * 1870–1904: non-interest-bearing deposits from Sveriges Riksbank (1931) minus the National Debt Office's giro balance (Riksdagens revisorer, 1870a) and the state's accounts at the Riksbank identified in the auditors' reports on the state finances (Riksdagens revisorer, 1870b). We count Statskontoret's cash, the cash of funds under its management, all balances stated as held *i/hos Riksbanken* on folio or giro accounts, pension funds' Riksbank and folio balances, and the postal savings bank's giro balance. Agencies whose cash location is not reported are included only from 1895, when they became obliged to hold their funds at the Riksbank. The general post of-

fice reports a single cash balance that combines its Riksbank balance with the working cash held at the post offices. Before 1895 we exclude it entirely, since there was no obligation to hold funds at the Riksbank and including it would make private deposits negative in most years. From 1895 we include only its cash in excess of the working cash needed at the post offices, which we set to its average cash holdings in 1885–1894 (about 1.6 million kronor), on the assumption that only surplus funds were transferred to the Riksbank. Statskontoret's book balance is corrected for the embezzlement by its cashier in 1887–1901.

    - 1905–1928: non-interest-bearing deposits from Sveriges Riksbank (1931) minus the giro accounts of state institutions from Sveriges Riksbank (1908) (1908-1930) (including the National Debt Office's giro account but excluding its special accounts opened during the 1907–08 crisis). For 1905–1907 we apply the state's share of giro deposits reported in Sveriges Riksbank (1908).
    - 1929–1939: Fregert (2014), *giroräkning för affärsbanker* + *giroräkning för andra insättare* (cheque accounts of commercial banks and of other depositors).

- Nominal GDP: Edvinsson (2014).
- Liabilities of the Riksbank: Fregert (2014).
- Notes: Fregert (2014).

Reliability. The retail deposit series is fairly reliable throughout: it is based on the Riksbank's reported interest-bearing deposits, which were offered specifically to non-bank customers (except in 1870–1872, when banks could also hold them, and a few other instances when state agencies report their holdings). From this total we deduct only explicitly reported state accounts and, in 1873, bank deposits. The series of non-interest-bearing deposits of banks and other private depositors rests on reported figures from 1905

onwards. Before 1905 it is an estimate: it is the residual of the Riksbank's total non-interest-bearing deposits after deducting state funds reconstructed account by account from the auditors' reports. Because this residual is small relative to total deposits, year-to-year changes of about ±1 million kronor lie within measurement error, mainly because of year-end timing differences between the agencies' books and the Riksbank's. Applied to 1905–1906, the method yields state funds about 7% below the yearbook figure, so the pre-1905 series likely overstates private deposits somewhat. The combined series of all private deposits inherits this uncertainty before 1905.

B.8 Switzerland

- Central bank deposits:
  - For 1931: Swiss National Bank (1932), p. 324.
  - SNB Archive:
    - For 1933–1936: *Die Gliederung der Girogelder* (Signature: 1.8.223).
    - For 1939: *Volkswirtschaftliche und Statistische Abteilung der Schweizerischen Nationalbank*, *Die Guthaben der Girokunden am Monatsende*, 22 September 1942. NB: numbers extracted from graph.
  - Notes and total liabilities: Swiss National Bank (2007).
- Nominal GDP: Stohr (2016).

B.9 United Kingdom

- Central bank deposits: Thomas and Dimsdale (2017).
- Nominal GDP: Thomas and Dimsdale (2017).